

A robust bi-objective location-routing model for providing emergency medical services

A location-
routing model
for EMS

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Received 17 November 2018

Revised 25 May 2019

26 September 2019

2 October 2019

2 December 2019

Accepted 4 December 2019

Abstract

Purpose – This paper addresses a location-routing problem (LRP) under uncertainty for providing emergency medical services (EMS) during disasters, which is formulated using a robust optimization (RO) approach. The objectives consist of minimizing relief time and the total cost including location costs and the cost of route coverage by the vehicles (ambulances and helicopters).

Design/methodology/approach – A shuffled frog leaping algorithm (SFLA) is developed to solve the problem and the performance is assessed using both the ϵ -constraint method and NSGA-II algorithm. For a more accurate validation of the proposed algorithm, the four indicators of dispersion measure (DM), mean ideal distance (MID), space measure (SM), and the number of Pareto solutions (NPS) are used.

Findings – The results obtained indicate the efficiency of the proposed algorithm within a proper computation time compared to the CPLEX solver as an exact method.

Research limitations/implications – In this study, the planning horizon is not considered in the model which can affect the value of parameters such as demand. Moreover, the uncertain nature of the other parameters such as traveling time is not incorporated into the model.

Practical implications – The outcomes of this research are helpful for decision-makers for the planning and management of casualty transportation under uncertain environment. The proposed algorithm can obtain acceptable solutions for real-world cases.

Originality/value – A novel robust mixed-integer linear programming (MILP) model is proposed to formulate the problem as a LRP. To solve the problem, two efficient metaheuristic algorithms were developed to determine the optimal values of objectives and decision variables.

Keywords Disaster management, Emergency medical service (EMS), Location-routing, Robust optimization, Shuffled frog leaping algorithm (SFLA)

Paper type Research paper

1. Introduction

Disastrous situations are always full of ambiguities in terms of their frequency and severity. Therefore, relieving the affected people needs effective planning in terms of the service



Journal of Humanitarian Logistics
and Supply Chain Management
Vol. 10 No. 3, 2020
pp. 285-319

© Emerald Publishing Limited
2042-6747

DOI 10.1108/JHLSCM-11-2018-0072

The author would like to thank the Editor-in-Chief and autonomous reviewers for their valuable comments and suggestions, which helped us to improve the paper.

resources to decrease injuries and losses. In a crisis, wasted time may change life to death (Altay, 2012).

The process of planning, management and the control of the flow of resources to provide services for people after the occurrence of disasters is known as emergency or relief logistics (Sheu, 2007; Kim and Jayakrishnan, 2010; Santos *et al.*, 2010). As the severity of disasters has been increased exponentially around the world since 1950 (Özdamar and Ertem, 2015), the importance of accurate planning with the consideration of its complexity and uncertain nature has become more than ever. In such a situation, there are challenges which make the problem highly complex. Challenges can relate to the planning to provide services for injured people, limited time and resources and the inability to estimate an accurate number of injured people.

In general, the humanitarian relief chains (HRCs) seek high speed when providing emergency requirements for injured people in order to minimize the number of injuries and mortality. This can be done through the efficient and effective allocation of limited resources (Tofighi *et al.*, 2016). Relief logistics include all of the processes of estimating, supplying, transporting, maintaining and distributing the commodities, equipment, services and all of the necessities to both injured people and relief groups. All of these necessities must be provided in the minimum possible time (proper time) to the designated locations (proper location) and in the required quantities for the intended people and groups (certain individuals) via the scientific and accurate method (proper method) (Özdamar *et al.*, 2004).

Three steps are considered for the response process, including (1) demand point collection, (2) demand classification based on severity or urgency and (3) solving the routing problem and sending the vehicles (Luca *et al.*, 2015). In this regard, the location and routing decisions related to relief distribution planning in a post-disaster situation play a key role in providing timely emergency services.

Barbarosoğlu *et al.* (2002) focused on modeling the helicopter relief operation and flight planning in the Turkish Army during a disaster and they indicated its role in planning within the operational areas and in limiting the service-providing time. Yi and Kumar (2007) solved the logistics problem in disaster relief operations. They proposed ant colony optimization (ACO) to provide an optimal dispatching planning of commodities to the distribution centers in the affected areas. Two main research studies on disaster relief and evacuation logistics were suggested by Yi and Ozdamar (2007) and Özdamar and Yi (2008). Their studies aimed to minimize the total service delay. An integer programming model was then proposed by Altay (2012) for the capability-based location-allocation problem considering multiple resources. Salman and Gul (2014) presented a novel multi-period mixed-integer linear programming (MILP) model for locating additional emergency units and transporting the casualties to hospitals by ambulance. They investigated a case study of the Istanbul earthquake.

Leiras *et al.* (2014) provided a comprehensive literature review on humanitarian logistics research including trends and challenges. Rath and Gutjahr (2014) proposed a routing-location model for relief in disastrous conditions. They expressed that intermediate warehouses should be constructed as temporary and outpatient hospitals in order to perform the relief steps and to meet the injured people's needs at a higher speed.

Halskau (2014) investigated marine helicopter routing through a hub location problem. The objective was to minimize the average number of casualties in emergency conditions. Ahmadi *et al.* (2015) presented a new structured method for solving the routing problem in disastrous conditions. According to their proposed approach, information was imported gradually over the time and it attempted to design the routes in such a way that minimal changes occurred in routes at the planning horizon time. This method was investigated in several numerical examples. Knyazkov *et al.* (2015) evaluated dynamic ambulance routing for transferring patients in St. Petersburg. In fact, the objective of the problem was to reduce the transferring times of injured parties to hospitals to save the patients' lives. This was accompanied by selecting an optimal route and a target hospital for an ambulance in a large

city, resulting in many complications. They considered dynamic features for an urban environment, including traffic flow, population density and hospital capabilities. This led to the determination of emergency medical services (EMS) quality. They investigated the problem through a case study in St. Petersburg and concluded that the traffic conditions had a significant effect on decision-making and that it might change the entire route as well as the target hospital. [Chen and Yu \(2016\)](#) investigated the problem of a temporary facility location for EMS during a disaster concerning the transportation and demand rate in this situation. [Maghfiroh et al. \(2018\)](#) investigated a dynamic vehicle routing problem (VRP) for the last mile distribution during a disaster. They developed a modified simulated annealing (SA) and variable neighborhood search (VNS) algorithm to solve the problem with the aim of minimizing the total traveling time. An improved shuffled frog leaping algorithm (SFLA) was proposed by [Duan et al. \(2018\)](#) to study the dynamic emergency vehicle dispatching problem. They considered response time, accident severity and accident time windows as the main factors to develop an emergency vehicle dispatching model.

[Zhang et al. \(2018\)](#) presented a sustainable multi-depot location-routing problem (LRP) considering uncertain information. They concurrently took into account the travel time, emergency relief costs and CO₂ emissions in their model via uncertainty theory. They sought to solve the problem using a hybrid genetic algorithm (GA) and uncertain simulations. An integrated location-inventory-routing problem (LIRP) was proposed for pre- and post-disaster management by [Tavana et al. \(2018\)](#). They considered a multi-echelon humanitarian logistics network to provide an appropriate flow of relief products. They solved the problem using an improved non-dominated sorting genetic algorithm II (NSGA-II) algorithm. According to the discussed literature up to this stage, our model considers refueling the vehicles and the use of helicopters together with ambulances.

In the field of special studies on robust optimization (RO) in relief logistics, [Bozorgi-Amiri et al. \(2011\)](#), for the first time, introduced a multi-objective robust stochastic programming approach for relief logistics under uncertain conditions. A dynamic multi-objective robust LRP for relief distribution was suggested by [Bozorgi-Amiri and Khorsi \(2016\)](#) considering uncertain demands, travel time and the cost parameters. They implemented the ϵ -constraint method to cope with the multi-objectiveness of the problem. They tested the applicability of their proposed approach on a real case study. A robust stochastic vehicle routing and scheduling method for bushfire emergency evacuation were investigated by [Shahparvari and Abbasi \(2017\)](#) related to an Australian case study. They took into account the road availability and disruptions in their proposed model and they developed a greedy solution method to solve the problem. [Veysmoradi et al. \(2018\)](#) introduced a multi-objective open LRP for relief distribution networks considering split delivery and multi-mode transportation under uncertain conditions. They formulated the problem as a mixed-integer non-linear programming (MINLP) model, and they provided a hybrid solution using RO and fuzzy multi-objective programming. [Liu et al. \(2018\)](#) developed a RO model for relief logistics planning considering uncertain demands and transportation time. They implemented their proposed model in a case study problem in a city of China that had recently suffered from an earthquake. They proposed optimal management policies using sensitivity analyses.

After reviewing the literature, it can be found that considering the concurrent transportation planning of ambulances and helicopters has not been investigated adequately. On the other hand, there has been a lack of attention to response time as the main objective against the total cost minimization. Time plays a critical role in the relief logistics system. More importantly, during a disaster, vehicles should not be confronted with fuel supply problems since this disrupts the relief operations and it increases the response time. This subject was not considered in most studies as well. Moreover, uncertainty is regarded as one of the most basic features of the relief logistics system which is studied in our problem using the RO technique.

Since the investigated problem is an extension of LRP and it is classified as an NP-hard problem (Ghaffari-Nasab *et al.*, 2013), it cannot be solved using exact methods within a reasonable computational time. For this purpose, many researchers have worked on different multi-objective algorithms such as greedy-search-based multi-objective GA (Chang *et al.*, 2014), multiobjective particle swarm optimization (Ghasemi *et al.*, 2019) and an NSGA-II (Vahdani *et al.*, 2018) in order to solve the multi-objective problems that are particular to relief logistics. In this research, a multi-objective algorithm for the SFLA is developed to solve the problem approximately. SFLA is known as a high-quality algorithm that is used to solve optimization problems (Rahimi-Vahed and Mirzaei, 2007; Niknam *et al.*, 2011; Arshi *et al.*, 2014). Furthermore, the efficiency of the algorithm is evaluated in comparison with the ϵ -constraint method and the NSGA-II for small- and large-sized problems respectively.

In other words, a robust multi-objective optimization model is essential to provide effective, efficient and equitable relief logistics. Efficient algorithms are also needed to find good quality solutions in the aftermath of the disaster. In this study, we focus on these requirements as the main contributions.

The remaining sections of this paper are organized as follows. Section 2 describes the problem and it presents the proposed robust mathematical model. The suggested solution approaches include the ϵ -constraint method, SFLA and NSGA-II as introduced in Section 3. Section 4 represents the computational results including the algorithm comparison, robust and deterministic problem comparison and sensitivity analysis. Finally, the concluding remarks and outlook of research are explained in Section 5.

2. Problem description

In this paper, a relief network is considered which includes fixed hospitals, potential temporary hospitals, transfer points, ambulance stations, helicopter stations, demand points and refueling points, and the number and location of each one are predetermined. This network is defined within the operational area where many injured people are waiting for relief. To simplify the problem, only two types of injured people are defined, namely, those who are treated at the site of the incident (green-type) and those who should be transferred to the hospital, due to the severity of their injury (red-type) (Tofighi *et al.*, 2016). Moreover, the potential points for facility establishment are predetermined in the relief network.

The overall structure of the proposed supply chain is as shown in Figure 1. In this figure, the red star points represent the red-type patients (acute care needed), green star points represent the green-type patients, blue circle points represent ambulance stations, blue triangle points represent transfer stations, blue hexagonal points represent potential temporary hospitals, blue square points represent fixed hospitals, blue rectangular points represent helicopter stations and blue dodecagon points represent gas stations. Since the landing and taking-off of the helicopters occur at pre-defined locations, we need to consider transfer stations in the network (Bozorgi-Amiri *et al.*, 2017). In fact, transferring injured people to the helicopter depot is done by an ambulance and then, transferring them to the hospital is performed by a helicopter.

In conclusion, our proposed model attempts to concurrently improve the efficiency, effectiveness and equity of the allocation of critical resources during the response phase. The measure of efficiency is denoted by the maximum time required by ambulances to leave disaster-affected points (Huang *et al.*, 2012). The measure of effectiveness can be represented by the speed and sufficiency of the deliveries (Huang *et al.*, 2012), which is studied by considering the assignment of helicopters to provide instantaneous services. Finally, the equity measure is addressed by the time window of each point to provide fair distribution among injured people (Gutjahr and Nolz, 2016). The arrival time of the ambulances at each point should occur within a given interval.

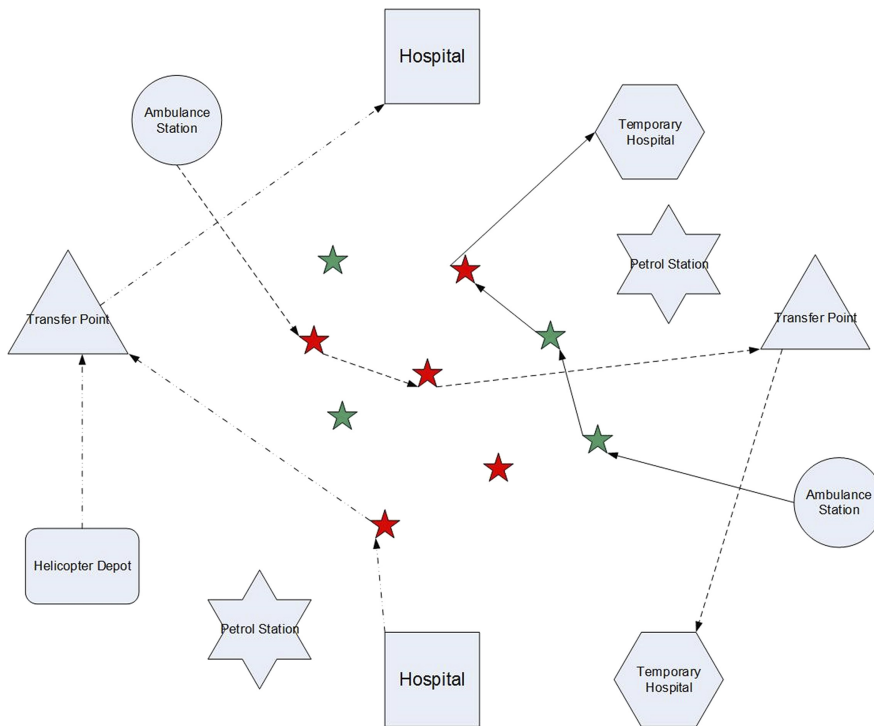


Figure 1.
Relief network
configuration

The main assumptions of the problem are as follows:

- (1) The distance between different points of the network is deterministic.
- (2) Ambulances and helicopters have limited capacity for providing services for patients.
- (3) Ambulances have a limitation of fuel consumption.
- (4) There are several potential points for the construction of temporary hospitals and transfer points.
- (5) Patients are divided into two groups of green and red.
- (6) Each ambulance can carry one or more red patients up to its capacity to the hospitals or transfer points.
- (7) Each ambulance can initially handle a green-type patient and then carry the red-type patient to the hospital or transfer point.
- (8) Hospitals have enough capacity to service all the red-type patients.
- (9) The number of patients after the incident is uncertain.
- (10) A hard time window is defined for each affected area (demand point).
- (11) Each helicopter should land at a transfer station and then transfer the patients to the hospitals.

(12) The time spent by helicopters is instantaneous.

The following symbols are used in this paper. Note that *min*, *lit* and *man* are the abbreviations of minute, liter and the number of people, respectively.

2.1 Sets and indices

I : Set of all points ($i, i', j, j' \in I$)
 H : Set of fixed hospitals ($H \subset I$)
 TH : Set of potential temporary hospitals ($TH \subset I$)
 TP : Set of transfer stations ($TP \subset I$)
 WS : Set of ambulance stations ($WS \subset I$)
 HS : Set of helicopter stations ($HS \subset I$)
 AA : Set of demand point ($AA \subset I$)
 FP : Set of gas stations ($FP \subset I$)
 K : Set of ambulances ($k \in K$)
 HC : Set of helicopters $h \in HC$

2.2 Parameters

t_{ij} (min): Travel time between the two points i and j by each of the ambulances
 QA_k (man): Capacity of the kth ambulance
 QH_h (man): Capacity of the hth helicopter
 DR_i (man): Number of red-type patients at the ith disaster-affected point, which is defined in an uncertainty range
 DR_i (man): Fluctuation rate of the number of red-type patients at the ith point
 DR_i (man): Estimated number of red-type patients at the ith point
 DG_i (man): Number of green-type patients at the ith point
 ST_i (min): Duration of providing services for green-type patients at the ith point
 SG_i (min): Duration of preparing the red-type patients for being transferred from the ith point
 FA_k (lit): Primary fuel of the kth ambulance
 FB_k (lit): Fuel of each ambulance after refueling at the gas station
 HP_{hi} : Binary parameter will be equal to 1 if the helicopter h is at the ith helicopter station at the starting time
 α (lit/min): Conversion factor of the elapsed time to the consumed fuel in ambulances
 cx_{ij} (\$): Cost of covering the i -to- j route by each ambulance
 cy_{ij} (\$): Cost of covering the i -to- j route by each helicopter
 fc_i (\$): Cost of constructing the transfer station i
 fd_i (\$): Cost of constructing the temporary hospital i
 E_i (min): Lower bound defined for the time window of point i
 L_i (min): Upper bound defined for the time window of point i

2.3 Decision variables

P_{ki}^S : Binary variable is equal to 1 if the ambulance k is at the ith station at the starting time
 P_{ki}^H : Binary variable is equal to 1 if the ambulance k is at the ith hospital at the starting time
 y_i : Binary variable is equal to 1 if the transfer station i is established
 y'_i : Binary variable is equal to 1 if the temporary hospital i is established
 x_{ijk} : Binary variable is equal to 1 if the ambulance k covers the route between points i and j
 z_{ijh} : Binary variable is equal to 1 if the helicopter h covers the route between points i and j
 δ_{ki} (lit): Residual fuel of the kth ambulance at the time of arriving at the ith point
 t_{ki}^{arrive} (min): Arrival time of the ambulance k to the ith point

t_{ki}^{left} (min): Time of leaving the disaster-affected point i by ambulance k

nx_{ki} (man): Number of red-type patients transferred from the i th disaster-affected point by the k th ambulance

rx_{ki} (man): Number of red-type patients delivered by ambulance k to the transfer station i

ar_{ki} (man): Number of red-type patients delivered by ambulance k to the hospital i

ag_{ki} (man): Number of green-type patients at the area i treated by ambulance k

x'_{ijh} (man): Number of red-type patients transferred by helicopter h from the transfer point i to the hospital j

Accordingly, the deterministic mathematical model is presented as follows:

$$\sum_{i \in WS} P_{ki}^S + \sum_{i \in H} P_{ki}^H = 1 \quad \forall k \in K \quad (1)$$

$$\sum_{j \in I} x_{ijk} \leq P_{ki}^S \quad \forall i \in WS, k \in K \quad (2)$$

$$\sum_{j \in I} x_{ijk} \leq P_{ki}^H \quad \forall i \in H, k \in K \quad (3)$$

$$\sum_{i \in I} x_{iik} = 0 \quad \forall k \in K \quad (4)$$

$$\sum_{i \in I} x_{ijk} - \sum_{i \in I} x_{jik} = 0 \quad \forall j \in I \quad (5)$$

$$\sum_{i \in I} x_{ij1k} + \sum_{i \in I} x_{ij2k} = 1 \quad \forall j \in TP, j' \in H \quad (6)$$

$$\delta_{ki} = FA_k P_{ki}^S \quad \forall i \in WS, k \in K \quad (7)$$

$$\delta_{ki} = FA_k P_{ki}^H \quad \forall i \in H, k \in K \quad (8)$$

$$\delta_{ki} - \alpha t_{ij} = \delta_{kj} + M(1 - x_{ijk}) \quad \forall i, j \in I, k \in K \quad (9)$$

$$\delta_{kj} = FB_k \sum_{i \in I} x_{ijk} \quad \forall j \in FP, k \in K \quad (10)$$

$$E_i \leq t_{kj}^{arrive} = t_{ki}^{left} + t_{ij}(1 - x_{ijk}) \leq L_i \quad \forall i, j \in AA, k \in K \quad (11)$$

$$\sum_{k \in K} nx_{ki} \geq DR_i \quad \forall i \in AA \quad (12)$$

$$\sum_{k \in K} rx_{ki} \leq My_i \quad \forall i \in TP \quad (13)$$

$$\sum_{i \in I} x_{ijk} \leq My_j \quad \forall j \in TP, k \in K \quad (14)$$

$$\sum_{i \in AA} nx_{ki} = rx_{kj} \sum_{i \in I} x_{ijk} \quad \forall j \in TP, k \in K \quad (15)$$

$$\sum_{i \in AA} nx_{ki} = ar_{kj} \sum_{i \in I} x_{ijk} \quad \forall j \in H \cup TH, k \in K \quad (16)$$

$$t_{ki}^{left} = t_{ki}^{arrive} + ST_i ag_{ki} \quad \forall i \in AA, k \in K \quad (17)$$

$$\sum_{k \in K} ag_{ki} = DG_i \quad \forall i \in AA \quad (18)$$

$$\sum_{k \in K} nx_{ki} \leq QA_k \quad \forall i \in AA \quad (19)$$

$$\sum_{i \in HS} z_{ijh} \leq y_j \quad \forall j \in TP, h \in HC \quad (20)$$

$$\sum_{i \in TP \cup HS} z_{ijh} - \sum_{i \in TP \cup H \cup TH} z_{jih} = 0 \quad \forall j \in TP, h \in HC \quad (21)$$

$$\sum_{i \in HS} \sum_{j \in H \cup TH} z_{ijh} = \sum_{i \in HS \cup TP} z_{ij'h} \quad \forall j' \in TP, h \in HC \quad (22)$$

$$\sum_{i \in TP} \sum_{j \in H \cup TH} x'_{ijh} \leq QH_h \quad \forall h \in HC \quad (24)$$

$$\sum_{h \in HC} \sum_{j \in H \cup TH} x'_{ijh} = \sum_{k \in K} rx_{ki} \quad \forall i \in TP \quad (25)$$

$$x'_{ijh} \leq M z_{ijh} \quad \forall i \in TP, j \in H \cup TH, h \in HC \quad (26)$$

$$\sum_{i \in TP} x'_{ijh} \leq M y'_j \quad \forall j \in TH, h \in HC \quad (27)$$

$$\sum_{i \in I} \sum_{h \in HC} z_{ijh} \leq M y'_j \quad \forall j \in TH \quad (28)$$

$$P^S_{ki}, P^H_{ki}, y_i, y'_i, x_{ijk}, z_{ijh} \in \{0, 1\} \quad \forall i, j \in I, k \in K, h \in HC \quad (29)$$

$$\delta_{ki}, t^{arrive}_{ki}, t^{left}_{ki} \geq 0; nx_{ki}, rx_{ki}, ar_{ki}, ag_{ki}, x'_{ijh} \in Z^+ \quad \forall i, j \in I, k \in K, h \in HC \quad (30)$$

The first objective function minimizes relief time, which includes minimization of the maximum time required by the ambulance to leave the disaster-affected points. The second objective function is related to the minimization of the costs, which includes minimization of the routing costs of vehicles and locating temporary hospitals and transfer stations.

Constraint (1) states that an ambulance is at one of the stations, or it is being used in one of the hospitals at the beginning of the time horizon. Constraint (2) states that each ambulance in any station must start its trip from the same point. Constraint (3) states that each ambulance in any hospital must start its trip from the same point. Constraint (4) states that a loop cannot be generated at any point. Constraint (5) states that an ambulance should leave the point that it previously entered. Constraint (6) states that the destination of each ambulance is the transfer stations or the hospitals. Constraints (7) and (8) determine the primary required fuel for ambulances. Constraint (9) calculates the amount of fuel consumption in the routes which are covered by each ambulance. Constraint (10) specifies the refueling of ambulances at the gas station. Constraint (11) determines the arrival time to and departure time from each of the affected points for the ambulances. Constraint (12) states that all the red-type patients must be transferred from the affected areas by ambulances. Constraint (13) states that if a transfer station is established, then the red-type patients can be transferred to it. Constraint (14) states that the ambulances can only travel to the established stations. Constraint (15) specifies the number of red-type patients at each transfer station. Constraint (16) determines the number of red-type patients that are directly transferred to the hospitals by ambulances. Constraint (17)

considers the departure time of each ambulance from the affected points equal to its arrival time plus the time of providing reliefs for the green-type patients. Constraint (18) states that all of the green-type patients should be treated. Constraint (19) represents the capacity limitation of ambulances. Constraint (20) states that helicopters are allowed to enter the transfer stations that were previously established. Constraint (21) states that if a helicopter enters a certain transfer station, it must leave there toward another transfer station or hospital. Constraint (22) states that only those helicopters are allowed to go to the hospital that previously stopped at one of the transfer stations. Constraint (23) represents the capacity limitation of the helicopters. Constraint (24) indicates that at each transfer station, all of the red-type patients brought by the ambulances should be transferred to the hospitals by helicopters. Constraint (25) indicates that a helicopter should land at a transfer station and then transfer the patients to the hospitals. Constraints (26) and (27) imply that the patients are only sent to the temporary hospitals which are already established. Constraint (28) states that the helicopters can only go to those temporary hospitals that are already established. Constraints (29) and (30) represent the type of decision variables.

2.4 Linearization of the mathematical model

In the proposed mathematical model, Constraints (15) and (16) cause the non-linearity of the proposed mathematical model. Using the definition of the variable, these constraints can be linearized and as a result, a complex integer programming model can be presented. Since the value of $\sum_{i \in I} x_{ijk}$ is equal to 0 or 1, the following conversions can be made (Tirkolaee *et al.*, 2017).

$$\sum_{i \in AA} nx_{ki} = rx_{kj} \sum_{i \in I} x_{ijk} \quad \forall j \in TP, k \in K \quad (31)$$

$$\sum_{i \in AA} nx_{ki} = va_{kj} \quad \forall j \in TP, k \in K \quad (32)$$

$$va_{kj} \leq rx_{kj} \quad \forall j \in TP, k \in K \quad (33)$$

$$va_{kj} \leq M \sum_{i \in I} x_{ijk} \quad \forall j \in TP, k \in K \quad (34)$$

$$va_{kj} \geq rx_{kj} - M \left(1 - \sum_{i \in I} x_{ijk} \right) \quad \forall j \in TP, k \in K \quad (35)$$

$$va_{kj} \in Z^+ \quad \forall j \in TP, k \in K \quad (36)$$

For Constraint (16), we have:

$$\sum_{i \in AA} nx_{ki} = hx_{kj} \sum_{i \in I} x_{ijk} \quad \forall j \in H \cup TH, k \in K \quad (37)$$

$$\sum_{i \in AA} nx_{ki} = vb_{kj} \quad \forall j \in H \cup TH, k \in K \quad (38)$$

$$vb_{kj} \leq hx_{kj} \quad \forall j \in H \cup TH, k \in K \quad (39)$$

$$vb_{kj} \leq M \sum_{i \in I} x_{ijk} \quad \forall j \in H \cup TH, k \in K \quad (40)$$

$$vb_{kj} \geq hx_{kj} - M \left(1 - \sum_{i \in I} x_{ijk} \right) \quad \forall j \in H \cup TH, k \in K \quad (41)$$

$$vb_{kj} \in Z^+ \quad \forall j \in H \cup TH, k \in K \quad (42)$$

2.5 Robust counterpart

RO is one of the most efficient approaches for investigating the effects of real-world uncertainty in a problem (Tirkolaee *et al.*, 2018, 2020; Golpîra and Tirkolaee, 2019). In this section, the robust mathematical model of the problem is presented based on the Bertsimas and Sim's (2004) model as follows:

The number of red-type patients at the i th point is indicated by DR_i . Suppose that the number of red-type at the disaster affected points is not clear and constant and might be afflicted by uncertainty based on the real-world conditions. To consider such conditions, an uncertainty interval is considered for this parameter; i.e. $[DR_i - \widehat{DR}_i, DR_i + \widehat{DR}_i]$ based on the robust approach of Bertsimas and Sim (2004). Following the interval uncertainty space, each of the uncertain DR_i is in the form of a symmetric and finite distance with DR_i as its center. Moreover, $\widehat{DR}_i$ is the fluctuation rate of this parameter, which is obtained from the equation $\widehat{DR}_i = \rho \overline{DR}_i$, and $\rho > 0$ is the level of uncertainty.

The objectives of the problem do not change and the added constraints are presented as follows:

$$\sum_{k \in K} nx_{ki} \geq DR_i \psi_i + zz_i \Gamma_i + \sum_{k \in K} rr_{ki} \quad \forall i \in AA \quad (43)$$

$$zz_i + rr_{ki} \geq \widehat{DR}_i E_i \quad \forall i \in AA, \forall k \in K \quad (44)$$

$$\psi_i = 1 \quad \forall i \in AA \quad (45)$$

$$-E_i \leq \psi_i \leq E_i \quad \forall i \in AA \quad (46)$$

$$rr_{ki}, E_i, zz_i \geq 0 \quad \forall i \in AA, \forall k \in K \quad (47)$$

Constraints (1) to (14)

Constraints (17) to (30)

Constraints (31) to (35)

Constraints (36) to (40).

Constraint (12) states that all of the red-type patients should be transferred from the affected areas by ambulances. After developing a robust model, this constraint transforms into Equation (41) (Bertsimas and Sim, 2004), and then Constraints (41)–(45) enter into the problem instead of Equation (12).

It should be noted that Γ_i in Equation (41), takes a value between 0 and the number of red-type patients at the i th point, which is equal to DR_i . If Γ_i is equal to DR_i , it indicates that an improper limit was selected for the parameter. This also implies the highest conservatism, while a value of zero indicates the lack of control over uncertainty. The decision-maker can consider a value for it regarding the conservatism level. Furthermore, since the parameter DR_i is the only right-side vector of Constraint (12) without being multiplied by any variable, variable ψ_i and constraint $\psi_i = 1, \forall i \in AA$ are considered conventionally to represent the structure of the robust model proposed by Bertsimas and Sim (2004).

3. Solution methods

In this section, the developed solution methods for this problem are described. Furthermore, in order to validate the methods, the validation of the solution method is done through randomly generated instances. Our proposed solution methods include the ε -constraint method and the SFLA. To solve the problem, ten breakpoints are regarded for each objective function, and accordingly, ten Pareto points are generated for each instance (Tirkolaee *et al.*, 2019a).

3.1 The epsilon constraint method (ϵ -constraint)

The ϵ -constraint technique is one of the well-known methods for dealing with multi-objective problems. This solves this type of problem by transferring all of the objective functions, except for one of them, into the constraints in each step (Ehrgott and Gandibleux, 2003; Bérubé *et al.*, 2009). One of the advantages of the ϵ -constraint method, compared to other exact methods such as the objectives' weighted sum method, is the ability to control the number of generated efficient solutions in the ϵ -constraint method, which is impossible in the weighted sum method (Mavrotas, 2009). By considering the values selected for ϵ_i , the problem is solved.

In the proposed ϵ -constraint method, the first objective function is considered to be the main objective and the second objective function is considered to be the second objective. The results obtained using the ϵ -constraint method are presented in the "Numerical Results" section.

3.2 Multi-objective shuffled frog leaping algorithm (MOSFLA)

SFLA is the developed version of the shuffled complex evolution algorithm (SCE or SCE-UA). This is an algorithm with a relatively long background in smart optimization (Eusuff *et al.*, 2006). SCE-UA is a combination of the evolutionary capabilities of GA with the random searching capability of the controlled random searching (CRS). Therefore, it can be classified among memetic algorithms.

By adding elitism and swarm intelligence capabilities to the SCE algorithm, the SFLA with several structural similarities is constructed. The similarity and relationship are so much that during the implementation of each of these algorithms, by applying a few changes, the computer program and implementation of each algorithm can be converted to the other.

In memetic algorithms, in contrast with GA, the attributes and capabilities are inherited to the children by parents. In memetic algorithms, each person (with respect to Lamarck's evolutionary theory) obtains useful attributes and characteristics by searching in their neighborhood (local search). This means that, in addition to the evolution of the population, the evolution also proceeds individually. Memetic algorithms are henceforth sometimes known as hybrid algorithms (HA) and local genetic algorithms (LGA).

The main steps of the algorithm in problem-solving are as follows:

Step 0: Selecting m and n , where m and n represent the number of groups (memplex) and the number of frogs in each group, respectively; therefore, the total population is calculated through $F = mn$.

Step 1: Generating primary random possible solutions in the number of F (primary population) and calculating the objective function of each solution.

Moreover, each solution consists of the following information:

- (1) Number of demand points allocated to each ambulance
- (2) Route of each ambulance
- (3) Location of each ambulance at the starting moment
- (4) Destination of each ambulance
- (5) Ambulances' arrival time to demand points
- (6) Ambulances' departure time from demand points
- (7) Gas stations that each ambulance should visit
- (8) Established temporary hospitals
- (9) Established transfer points

- (10) Demand collected at each transfer point
- (11) Number of transfer points allocated to each helicopter
- (12) Route of each helicopter
- (13) Location of each helicopter at any starting moment (this is additional information and is given for ease of calculation)
- (14) Destination of each helicopter
- (15) Travel costs of each ambulance
- (16) Establishment costs of temporary hospitals
- (17) Establishment costs of transfer points
- (18) Travel costs of each helicopter

Step 2: Sort the frogs based on their fitness. Store the best solution (F_g) in the total population.

Step 3: Put $it = 1$ ($it = 1, 2, \dots, N$), where N represents the maximum search iterations. Divide the frogs into the groups (M_k) as follows:

$$M^k = \{F_j | j = k + m(j-1), j = 1, 2, \dots, n\} \quad k = 1, 2, \dots, m \quad (48)$$

For example, in case of dividing 30 frogs into six groups with five frogs, the indices 1, 7, 13, 19 and 25 are located in the first group, and the indices 4, 10, 16, 22 and 28 are located in the fourth group.

Step 4: Frog locating algorithm (FLA):

Substep 4.1: Put $im = 1$ ($im = 1, 2, \dots, m$).

Substep 4.2: Put $in = 1$ ($in = 1, 2, \dots, b$) (counter of the number of search iterations in each group).

Substep 4.3: Select q frogs from among the frogs of the im group randomly with the following probability (sub-grouping); $p_j = 2(n+1-j)/n(n+1)$, ($j = 1, \dots, n$).

Substep 4.4: Put $k = 1$ ($k = 1, 2, \dots, q$).

Substep 4.5: Select an improvement algorithm (among existing algorithms) randomly with probabilities of (0.2, 0.2, 0.4 and 0.2).

Substep 4.6: Implement the selected improvement algorithm on the solution of the k th frog and obtain a new solution. If the new possible solution is better than the solution of the k th frog, then replace the new solution for the worst solution in the sub-group and then go to Substep 4.8. Otherwise (if the new solution is impossible, or if it is not better than the solution of the k th frog), select another improvement algorithm and re-implement this step.

If no new possible solution is obtained with all the 4 improvement algorithms, go to Substep 4.7.

Substep 4.7: Replace a new random solution for the worst solution in the subgroup and then go to Substep 4.8.

Substep 4.8: Put $k = k + 1$ and if $k \leq q$, go back to Substep 4.4; otherwise, go to Substep 4.9.

Substep 4.9: Put $in = in + 1$ and if $in \leq b$, go back to Substep 4.2; otherwise, go to Substep 4.10.

Substep 4.10: Put $im = im + 1$ and if $im \leq m$, go back to Substep 4.1; otherwise, go to Step 5.

Step 5: Re-sort the existing solutions based on their fitness. Update the best solution in the total population (Fg). Set $it = it + 1$ and if $it \leq N$, go to Step 3; otherwise, stop.

3.3 General structure for the generation algorithm of primary random solutions

The random solutions are generated through a heuristic method as follows:

Step 1: Generating a random formation

In this step, all the demand points are divided among the ambulances. In other words, when we have D demand points and $|K|$ ambulances, then the total number of formations will be equal to $D|K|$ and thus in this step, we select a formation randomly.

For example, to divide 30 demand points among 4 ambulances, the random formations of 8, 2, 15 and 5 is obtained in this step.

Step 2: Allocating the demand points to ambulances randomly based on the formations of *Step 1* and determining the route of each ambulance

In this step, first, the demand points are sorted randomly. Then, they are allocated to the ambulances based on the formations of step 1. For the previous example, after random sorting of the demand points, the first eight points are allocated to ambulance 1, 2, next points to ambulance 2, 15, next points to ambulance three and, the last 5 demand points to ambulance 4.

Step 3: Checking the capacity of the ambulances

In the previous step, the route of each ambulance was obtained. Consequently, the capacity of the ambulances should be investigated. If the capacity of an ambulance allows going to the allocated demand points, we go to the next step. Otherwise, we return to the previous step.

Step 4: Checking the time window constraint of demand points and calculating the time and cost

In this step, by calculating the arrival time of each ambulance to each demand point as well as its departure time from that point, the time window constraint is investigated. Also, residual fuel of the ambulances is taken into account and if necessary, the time of going to the gas station for refueling will be taken into consideration as well.

Furthermore, in this step, the primary location of the ambulances (an ambulance station or hospital) is also calculated and determined; moreover, the destination of the ambulances is determined randomly in this step.

In this step, if the time window constraint is not met for a demand point, we go back to Step 1. Otherwise, we will go to Step 5.

Step 5: Generating the route of helicopters

In the previous step, the destination of the ambulances was determined where some of the ambulances delivered their patients to the transfer points; therefore, the route of helicopters should be determined to take these patients to the (fixed or temporary) hospitals. For this

purpose, firstly, the demand collected at the transfer points is randomly allocated to the helicopters regarding their capacity. This procedure is performed almost similar to the solution of the transportation problem, but randomly, not based on the transportation cost. After allocating the demand collected at the transfer points among the helicopters, the cost of traveling through the routes (from the primary location of the helicopter to the transfer point, between transfer points, from transfer points to fix or temporary hospitals) is also calculated.

3.4 General structure of improvement algorithms

3.4.1 Improvement algorithm 1: changing the primary random formation of dividing the demand points among the ambulances (Step 1: random solution generation algorithm). Two ambulances are selected randomly. In the primary random formation, a demand point is taken from an ambulance and then it is allocated to another ambulance. Then, this change is applied on the route of those two ambulances as well.

Such changes might result in altering the capacity constraint of these two ambulances; thus, the capacity constraint is re-checked for these two ambulances.

Moreover, the arrival and departure time from the demand points, as well as the traveling costs, are re-calculated for these two ambulances; moreover, the time window constraint is re-investigated.

Finally, helicopters are re-routed as well.

3.4.2 Improvement algorithm 2: changing the route of an ambulance. An ambulance is selected randomly and then two or more demand points are displaced in the route. For example, if the route of the selected ambulance is:

8, 14, 5 and 15

Random displacement of 3 points: The points are displaced with random order of 1, 4, 3.

(1) Point 3 is replaced by point 1.

(2) Point 4 is replaced by point 3.

(3) Point 1 is replaced by point 4.

5, 14, 15 and 8

After applying this change, the arrival and departure time are recalculated for this ambulance as well as the costs; furthermore, since no change is made in the destination of the ambulances, there is no need for re-routing the helicopters.

3.4.3 Improvement algorithm 3: changing the destination of the ambulances. In this algorithm, the destination of all ambulances is determined randomly again and the previous destinations are eliminated. Then, due to the probable changes in the amount of the demand collected at the transfer points, the helicopters are re-routed as well.

3.4.4 Improvement algorithm 4: Re-routing of helicopters. In this algorithm, the demand collected at the transfer points is randomly divided among the helicopters and the helicopters are re-routed.

Flowchart of the proposed SFLA algorithm is shown in [Figure 2](#).

3.5 NSGA-II algorithm

[Srinivas and Deb \(1994\)](#) introduced the NSGA. They divided the evolutionary group into several levels according to a dominance relation for selection and solution. An operational NSGA scale was then optimized by [Deb et al. \(2002\)](#) to apply the elite mechanism instead of sharing coefficient of the density function. This algorithm is called NSGA-II. The suggested pseudo-code of NSGA-II in this research is given in [Figure 3](#).

Solution representation and the main mechanism of the proposed NSGA-II are discussed in the following.

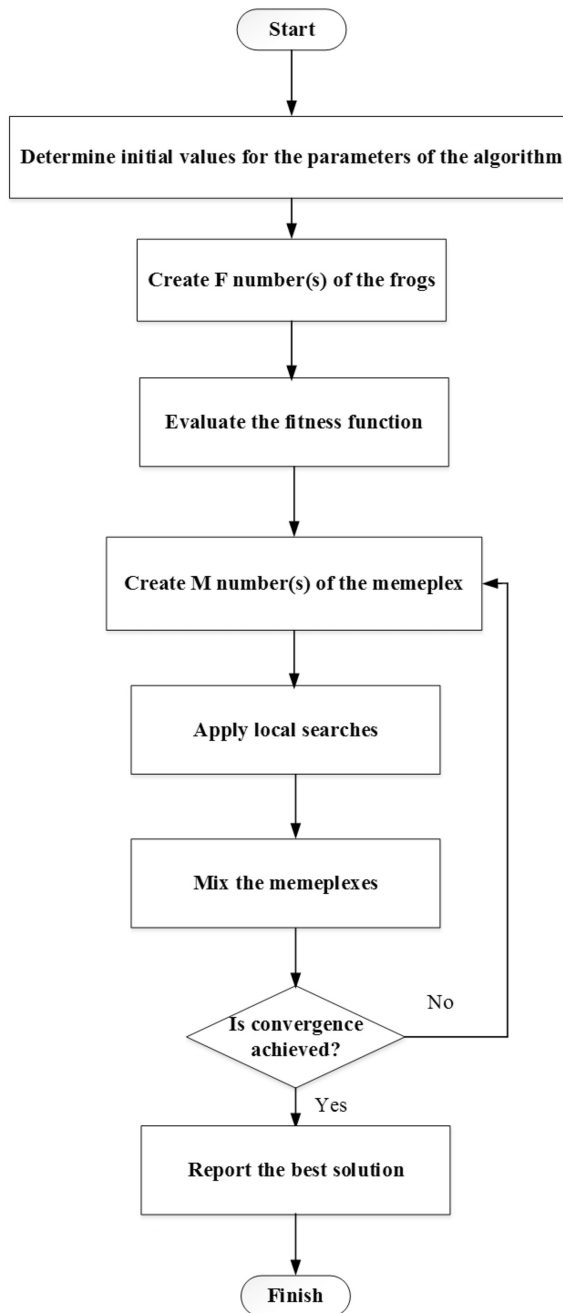


Figure 2.
Flowchart of SFLA
algorithm

Figure 3.
Pseudo-code of NSGA-II (Assunção *et al.*, 2013)

Procedure NSGA-II

Input: N' , g , $f_k(X)$ ▷ N' members evolved g generations to solve $f_k(X)$

- 1 Initialize Population $\mathbb{P}'$;
- 2 Generate random population - size N' ;
- 3 Evaluate Objectives Values;
- 4 Assign Rank (level) based on Pareto - *sort*;
- 5 Generate Child Population;
- 6 Binary Tournament Selection;
- 7 Recombination and Mutation;
- 8 **for** $i = 1$ **to** g **do**
- 9 **for each** Parent and Child in Population **do**
- 10 Assign Rank (level) based on Pareto - *sort*;
- 11 Generate sets of nondominated solutions;
- 12 Determine Crowding distance;
- 13 Loop (inside) by adding solutions to next generation starting from the *first* front until N' individuals;
- 14 **end**
- 15 Select points on the lower front with high crowding distance;
- 16 Create next generation;
- 17 Binary Tournament Selection;
- 18 Recombination and Mutation;
- 19 **end**

3.5.1 Representation of the solution. In the proposed algorithms, to demonstrate an initial solution, a six-row matrix is used, such that all of its entries are 0 and 1. Figure 4 shows the defined string for the GA.

In Figure 4, in the first row of the matrix, there are $|K|*|I|$ columns and if the ambulance k is at the i th station at the starting moment, its relevant entry in this row takes the value of 1. In the second row of the matrix, there are $|K|*|I|$ columns and if the ambulance k is at the i th hospital at the starting moment, then its relevant entry in this row takes the value of 1.

In the third row of the matrix, there are $|I|$ columns and if the transfer station i is established, then the i th entry of this row takes the value of 1. The fourth row of the matrix has I columns and if the temporary hospital j is established, then the i th entry of this row takes the value of 1. The fifth row of the matrix has $|I|*|J|*|K|$ columns and the ijk th entry of this row is equal to 1 when the ambulance k travels the route between i and j . Similarly, the fifth row of the matrix has $|I|*|J|*|H|$ columns and the ijh th entry of this row is equal to 1 when the helicopter h travels the route between i and j .

In addition, for crossover operator in the NSGA-II, a two-point crossover operator is used, while for mutation operator, a single-point mutation operator is used.

4. Computational analysis

4.1 Instance generation

In this section, in order to analyze the validation of the presented mathematical model, several instances of the problem are generated randomly in different sizes. The information of each instance is as shown in Table 1. In order to generate instances in a small size (1st instance), medium size (2nd and 3rd instances) and large size (4th instance) 2D space, n nodes are considered such as the nodes of the fixed hospitals, nodes of potential temporary hospitals, nodes of transfer stations, nodes of ambulance stations, nodes of helicopter stations, demand

Figure 4.
Demonstration of the solution as a chromosome for GA

$k_1 i_1$	$k_1 i_2$	$k_1 i_3$	...	$k_1 i_l$	$k_2 i_1$	$k_2 i_2$	$k_2 i_3$	...	$k_2 i_l$	$k_K i_1$	...	$k_K i_l$
$k_1 i_1$	$k_1 i_2$	$k_1 i_3$	...	$k_1 i_l$	$k_2 i_1$	$k_2 i_2$	$k_2 i_3$	...	$k_2 i_l$	$k_K i_1$	...	$k_K i_l$
i_1	i_2	i_3	i_4	i_5	i_6	i_7	i_8	...	...	...	i_{l-1}	i_l
j_1	j_2	j_3	j_4	j_5	j_6	j_7	j_8	...	...	...	j_{l-1}	j_l
$i_1 j_1 k_1$	$i_1 j_1 k_2$	...	$i_1 j_1 k_K$	...	$i_1 j_2 k_K$	...	$i_1 j_3 k_K$	...	...	$i_1 j_l k_1$	...	$i_1 j_l k_K$
$i_1 j_1 h_1$	$i_1 j_1 h_2$	...	$i_1 j_1 h_H$	...	$i_1 j_2 h_H$	...	$i_1 j_3 h_H$	...	...	$i_1 j_l h_1$	...	$i_1 j_l h_H$

nodes and nodes of gas station points. By considering the different values for them, the instances are generated. In fact, different sized instances can be generated considering different sizes for the relief network (see Table 1). In each random instance, a unique uniform distribution is used which is shown in Table 2. Most of these values are adapted from previous research such as that by Najafi *et al.* (2013). Different values can yield different results and the impact of the parameters' change can be studied using sensitivity analysis. In each random instance, the value of Γ_i is equal to half of the total number of red-type injured people rounded upwardly and calculations are performed for the uncertainty levels of 0.2, 0.4 and 0.5. These values are selected because the maximum uncertainty level of 50% is probable in relief logistics (Zokaei *et al.*, 2016).

To solve the ϵ -constraint solution method, the GAMS (V-24.1) software was used and the algorithm was encoded using the MATLAB (V-16.0) software. The problem is solved separately in order to determine the Pareto points based on each objective function. It should be noted that all runs of the problem in the GAMS software are performed considering a time limitation of 3,600 s (Bozorgi-Amiri *et al.*, 2011). This time limitation is only set to test the performance of the proposed algorithms in small-sized instances. In fact, after executing the 3600-s time limitation, the best-obtained solution is reported which might be the overall optimal or local optimal solution. It should be noted that 4th instance in the ϵ -constraint method does not yield a solution within this time limitation. This instance can be solved only by the SFLA algorithm.

Instance	I	K	HC	AA	H	TH	TP	WS	HS	FP
1	10	4	1	4	1	1	1	1	1	1
2	25	10	2	12	3	3	2	2	1	2
3	45	15	4	21	5	5	5	3	3	3
4	100	20	10	57	8	10	5	5	5	10

Table 1.
Randomly generated
instances

Parameters	Values
t_{ij} (min)	Uniform(1,10)
QA_k (man)	8
QH_h (man)	50
UT_k (min)	5,000
$\widehat{DR}_i$ (man)	Uniform(2,5)
DC_i (man)	Uniform(9,15)
SG_i (min)	Uniform(1,2)
ST_i (min)	Uniform(5,10)
$F1_k$ (lit)	100
$F2_k$ (lit)	150
E_i (min)	1
L_i (min)	Uniform(1,000,2,000)
HP_{hi}	1
α (lit/min)	0.8
cx_{ij} (\$)	Uniform(10,15)
cy_{ij} (\$)	Uniform(30,45)
f_i (\$)	10,000
$f1_i$ (\$)	12,000
ρ	(0.2, 0.4, 0.5)
$\widehat{DR}_i$ (man)	$\rho \widehat{DR}_i$
Γ_i	$0.5 \widehat{DR}_i$

Table 2.
Values of parameters

In Table 1, from left to right, the columns represent the rank of the generated random instances, the number of nodes in the network, the number of available ambulances, the number of available helicopters, the number of disaster-affected points within the network, the number of fixed hospitals, the number of potential temporary hospitals, the number of transfer stations, the number of ambulance stations, the number of helicopter stations and the number of gas stations respectively.

5. Results

The results are reported for various levels of uncertainty in Tables 3–5 respectively. In these tables, the value reported for each objective function is obtained by solving the instances with that specific single objective. The gap reported in Tables 3–5 is calculated as follows:

$$\begin{aligned} \text{Gap}(\text{Min f1}) &= [\text{Obj.1}(\text{Min f2}) - \text{Obj.1}(\text{Min f1})]/\text{Obj1}(\text{Min f1}), \\ \text{Gap}(\text{Min f2}) &= [\text{Obj.2}(\text{Min f2}) - \text{Obj.2}(\text{Min f1})]/\text{Obj2}(\text{Min f1}). \end{aligned} \tag{49}$$

Then, with regard to the ϵ -constraint method for different levels of uncertainty, the values of the epsilons are determined for the second objective, which are presented in Table 6 for the different breakpoints. In Table 6, each instance is solved for the ten breakpoints and three levels of uncertainty so then the values of ϵ_2 are obtained.

As seen in Table 6, the uncertainty level has a direct impact on the value of the objectives. It demonstrates that higher uncertainty levels lead to higher (worse) values for the objectives by keeping in mind the feasibility of the solutions. The main reason is the presence of uncertainty in the number of injured people which requires more resources and facilities for transportation planning. Consequently, the service time increases as well.

Table 3.
Optimal values of
objective functions for
separate solutions for
uncertainty level of 0.2

Instance	Type of objectives	Obj. 1 (min)	Obj. 2 (\$)	Gap (%)	Solving time (sec)	Number of established transfer stations	Number of established temporary hospitals
1	Min f1	23.644	22380.98	430.959	6.35	1	1
	Min f2	125.540	22172.316	0.941	2.368	1	1
2	Min f1	19.264	57268.040	656.105	153.51	3	3
	Min f2	145.656	56387.996	1.561	145.892	2	3
3	Min f1	14.770	66632.308	912.390	2459.18	4	3
	Min f2	149.530	66143.446	0.739	2168.06	3	3

Table 4.
Optimal values of
objective functions for
separate solutions for
uncertainty level of 0.4

Instance	Type of objectives	Obj. 1 (min)	Obj. 2 (\$)	Gap (%)	Solving time (sec)	Number of established transfer stations	Number of established temporary hospitals
1	Min f1	29.823	34256.232	341.334	16.864	1	2
	Min f2	131.619	22172.316	54.500	10.18	1	1
2	Min f1	20.865	57597.704	616.995	895.03	3	3
	Min f2	149.601	56498.021	1.946	606.65	2	3
3	Min f1	20.569	66713.066	629.447	3,600	4	3
	Min f2	150.040	66228.240	0.732	3,600	3	3

The Pareto fronts of the proposed 1st-4th instances are presented in Figures 5–8 for the uncertainty level of 0.5. The 1st and 2nd instances are solved using the exact method, while the 3rd and 4th instances are solved using the SFLA algorithm.

According to Figures 5–8, different instances have different Pareto fronts with different dispersion rates. The obtained trade-offs indicate that the objectives interact reversely. In other words, by increasing one of them, the other decreases. Moreover, the dispersion of the

Table 5.
Optimal values of objective functions for separate solutions for uncertainty level of 0.5

Instance	Type of objectives	Obj. 1 (min)	Obj. 2 (\$)	Gap (%)	Solving time (sec)	Number of established transfer stations	Number of established temporary hospitals
1	Min f1	32.056	34427.035	320.074	22.982	1	2
	Min f2	134.659	32202.615	6.908	14.63	2	1
2	Min f1	20.865	57602.528	635.897	1395.56	2	3
	Min f2	153.545	56632.021	1.714	906.35	2	3
3	Min f1	24.832	66883.973	528.222	3,600	4	4
	Min f2	156.001	66419.24	0.700	3,600	3	4

Table 6.
Epsilon values of the second objective function (ϵ_2) for different uncertainty levels

Instance	Eakpoints	Epsilon value		
		Uncertainty level of 0.2	Uncertainty level of 0.4	Uncertainty level of 0.5
1	1	22193.18	23380.71	32425.06
2		476	56607.99	56729.07
3		66192.33	66276.72	66465.71
1	2	22214.05	24589.1	32647.5
2		56564	56717.96	56826.12
3		66241.22	66325.21	66512.19
1	3	22234.92	25797.49	32869.94
2		56652.01	56827.93	56923.17
3		66290.1	66373.69	66558.66
1	4	22255.78	27005.88	33092.38
2		56740.01	56937.89	57020.22
3		66338.99	66422.17	66605.13
1	5	22276.65	28214.27	33314.83
2		56828.02	57047.86	57117.27
3		66387.88	66470.65	66651.61
1	6	22297.51	29422.67	33537.27
2		56916.02	57157.83	57214.33
3		66436.76	66519.14	66698.08
1	7	22318.38	30631.06	33759.71
2		57004.03	57267.8	57311.38
3		66485.65	66567.62	66744.55
1	8	22339.25	31839.45	33982.15
2		57092.03	57377.77	57408.43
3		66534.54	66616.1	66791.03
1	9	22360.11	33047.84	34204.59
2		57180.04	57487.74	57505.48
3		66583.42	66664.58	66837.5
1	10	22380.98	34256.23	34427.04
2		57268.04	57597.7	57602.53
3		66632.31	66713.07	66883.97

Figure 5.
Pareto front obtained
from solving the first
instance by
 ϵ -constraint method
for uncertainty level
of 0.5

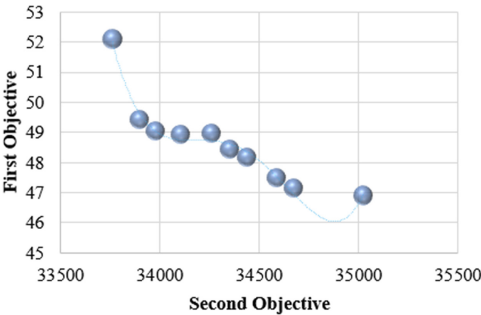


Figure 6.
Pareto front obtained
from solving the
second instance by
 ϵ -constraint method
for uncertainty level of 0.5

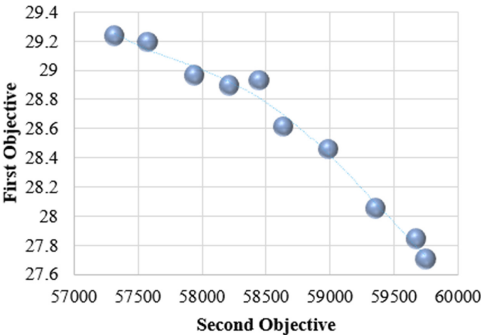
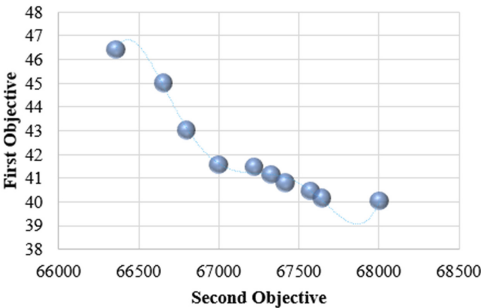


Figure 7.
Pareto front obtained
from solving the third
instance using SFLA
method for uncertainty
level of 0.5



Pareto fronts in each instance is different. To study the trade-offs and to choose the best values of these points, the reference point approach is implemented.

5.1 Optimal Pareto solution

In this subsection, an efficient approach is introduced to choose the best Pareto solutions. The selected Pareto solution can be investigated and implemented as part of the optimal planning of the relief logistics system. [Deb and Sundar \(2006\)](#) presented a reference point approach in the context of multi-objective problems in order to determine the best point or Pareto point. This method can be applied by either weighing the objectives or not. The main idea of this method is to highlight the solutions that are close to the reference points.

The normalized Euclidean distance (dev_j) between the solutions of the best non-dominated level and the reference point is calculated using [Equation \(48\)](#). Each solution with a lower value of dev_j has a higher priority for the decision-maker.

$$dev_j = \sqrt{\sum_{i=1}^N w_i \left(\frac{f_i - \bar{z}_i}{f_i^{\max} - f_i^{\min}} \right)^2} \quad (50)$$

In Equation (47), w_i is the weight of objective i in each solution, f_i is the value of objective i , $\bar{z}_i$ is the value of reference objective, and $f_i^{\max}$ and $f_i^{\min}$ are the maximum and minimum values for objective i respectively. The values of the parameters related to the determination of the best Pareto solution are presented in Table 7.

The results obtained from solving three instances are presented as the best Pareto solution at the break points and these results are compared with those of the meta-heuristic algorithm. Tables 8 and 9 represent the results obtained from both the ε -constraint method and the

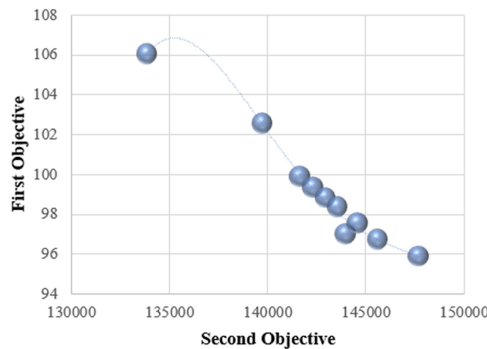


Figure 8.
Pareto front obtained
from solving the fourth
instance using SFLA
method for uncertainty
level of 0.5

Parameters	Values
w_1	0.7
w_2	0.3
$\bar{z}_1$ (\$)	$f_1^{\min}$
$\bar{z}_2$ (\$)	$f_2^{\min}$
$f_1^{\max}$ (\$)	Highest obtained value for the first objective among ten Pareto solutions in each problem
$f_2^{\max}$ (\$)	Highest obtained value for the second objective among ten Pareto solutions in each problem
$f_1^{\min}$ (\$)	Lowest obtained value for the first objective among ten Pareto solutions in each problem
$f_2^{\min}$ (\$)	Lowest obtained value for the second objective among ten Pareto solutions in each problem

Table 7.
Values of parameters
defined for the method

Instance	Uncertainty level of 0.2		GAMS solution		Uncertainty level of 0.5		Average run time (sec)
	Obj. 1(min)	Obj. 2 (\$)	Obj. 1(min)	Obj. 2 (\$)	Obj. 1(min)	Obj. 2 (\$)	
1	29,036	22235.126	63,044	23280.013	52,103	33759.71	33.56
2	39,264	57023.654	43.25	57007.704	30,774	57315.608	684.01
3	62.89	66410.228	32.569	66664.58	44.362	66651.61	3,600
4	—	—	—	—	—	—	3,600

Table 8.
Values of the objectives
obtained by the
 ε -constraint method
using the exact method

SFLA algorithm. In these tables, the values of the first and second objectives as well as the average solution time are presented for the different levels of uncertainty.

The results obtained from solving the problem through the exact method are presented in Figure 9 in different sizes for the different levels of uncertainty.

As shown in Figure 9, the value of the second objective increases from the small to large instances, meaning that increasing the size of the problem leads to the increase of the second objective. Since the first objective represents the minimization of the ambulances' maximum departure time from the affected points, it may be reduced by using a higher number of ambulances. However, there is a trade-off between these total cost increases and the number of injured people. The results of the SFLA algorithm are presented as follows.

Table 9.

Values of objectives obtained using SFLA algorithm

Instance	SFLA solution Uncertainty level of 0.2		Uncertainty level of 0.4		Uncertainty level of 0.5		Average run time (sec)
	Obj. 1(min)	Obj. 2 (\$)	Obj. 1(min)	Obj. 2 (\$)	Obj. 1(min)	Obj. 2 (\$)	
1	31.15	22813.2393	67.16	23975.8526	53.54	34420.7251	15.26
2	56.2	58184.0854	49.23	58347.385	29.95	59865.7232	46.15
3	61.05	64792.4749	32.01	66571.2496	41.66	68745.1481	147.28
4	125.16	121084.177	126.05	123583.966	106.05	133844.741	249.13

Figure 9.

Results obtained from solving different instances considering different levels of uncertainty using the exact method

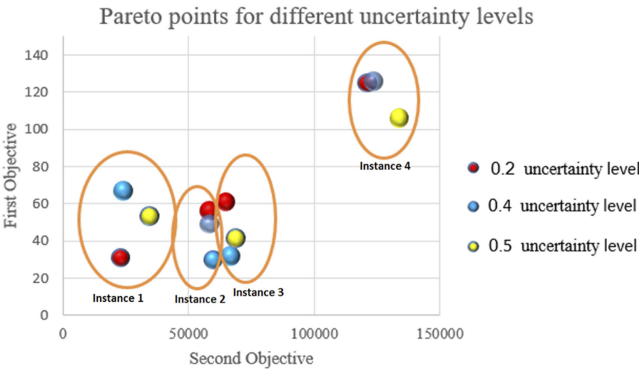
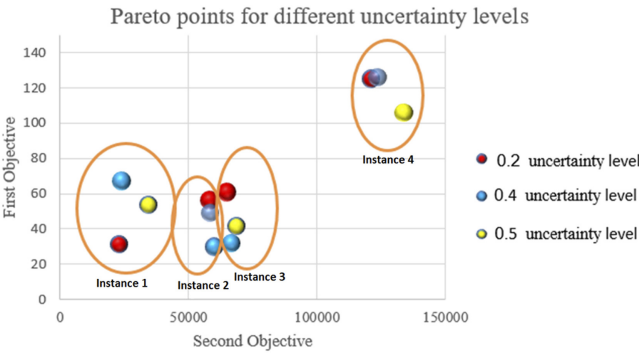


Figure 10.

Results obtained from solving different instances considering different levels of uncertainty using SFLA algorithm



The results obtained from solving the problem using the SFLA algorithm in different size for different levels of uncertainty are presented in Figure 10.

After reviewing the results in Tables 8 and 9 as well as in Figures 9 and 10, the results of the exact method and SFLA algorithm are close to each other and the proposed algorithm has a high performance. Furthermore, it yields better results for the uncertainty levels of 0.2 and 0.4 in the 3rd instance than in the applied exact method. The results of the NSGA-II algorithm are presented as follows.

According to Figure 11 and Table 10, the results of the NSGA-II algorithm are highly similar to the exact method and SFLA algorithm. Nevertheless, in order to specify the proximity of our meta-heuristic algorithms to the exact method and to evaluate their performance, several well-known indicators of the multi-objective algorithms are calculated in the following.

5.2 Performance of proposed meta-heuristic algorithms

In this section, the results are compared with those of the NSGA-II and SFLA algorithms and then the algorithms were validated. In fact, results of the NSGA-II and SFLA algorithms in instances with small and medium sizes are compared to the exact method by the CPLEX solver of GAMS software (Tirkolaee et al., 2020). Since it is not logical to solve the large-sized instances with exact methods, they were solved using meta-heuristic algorithms.

For a more accurate validation of the proposed algorithms and also to compare their performance, the indicators of the multi-objective algorithms were used. For this purpose, the four indicators of dispersion measurement (DM), mean ideal distance (MID), space measure (SM) and the number of Pareto solutions (NPS) were calculated and then the performance of the proposed algorithms was assessed. Higher values of DM and NPS show the higher ability

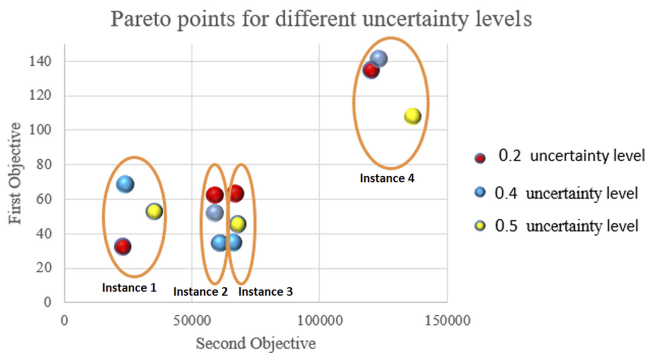


Figure 11. Results obtained from solving different instances considering different levels of uncertainty using NSGA-II algorithm

Instance	NSGA-II solution						Average run time (sec)
	Uncertainty level of 0.2		Uncertainty level of 0.4		Uncertainty level of 0.5		
	Obj. 1(min)	Obj. 2 (\$)	Obj. 1(min)	Obj. 2 (\$)	Obj. 1(min)	Obj. 2 (\$)	
1	32.57	22791.15	68.85	23845.15	52.95	35152.87	11.15
2	62.35	58953.35	51.84	58931.89	34.45	60854.55	28.84
3	63.14	66932.41	35.12	65952.74	45.57	67954.21	113.95
4	134.95	120169.5	141.58	122984.5	107.96	136514.9	180.51

Table 10. Objective values obtained using NSGA-II algorithm

of the algorithm to find the Pareto front. However, this is true for MID and SM with their lower values (Sadeghi and Niaki, 2015).

As shown in Tables A1–A3 (Appendix), the meta-heuristic algorithms used in this research perform very closely to the exact algorithm and their performance charts are very similar. Thus they are appropriately efficient for finding near-optimal solutions. Accordingly, due to such an appropriate performance, they can be used to solve the large-sized instances that cannot be solved by the ϵ -constraint method.

More specifically, in terms of the diversity criterion, the SFLA algorithm has a much better performance than the NSGA-II algorithm. This means that the SFLA algorithm is capable of generating Pareto fronts that include a wider range of possible solutions. However, the SFLA algorithm has functional fluctuations in terms of the diversity criterion. Thus, it does not have a better performance than the NSGA-II algorithm in the 1st instance with an uncertainty level of 0.5. It works so much better in the other instances. On this basis, disregarding the exceptions, it can be concluded that the SFLA algorithm exhibits a higher performance in this index.

In the criterion of distance from an ideal point, the SFLA algorithm has a better performance. This means that in this algorithm, the distance from the ideal solution is less than that of the other algorithms. In contrast to the diversity index, there is no exception in the index of distance from the ideal point and the SFLA algorithm has a better performance.

Furthermore, based on the distance criterion, the SFLA algorithm represents a better performance and it generates more approximate solutions. Thus it can be said that it has a more uniform front compared to the NSGA-II algorithm. Besides, the SFLA algorithm always succeeds to find more Pareto solutions. In fact, the main idea of implementing the NSGA-II algorithm is to test the performance of the SFLA algorithm in large-sized instances.

For a general comparison of the algorithms, the simple additive weighting (SAW) method is used (Zitzler *et al.*, 2000). Accordingly, all of the values obtained for the indicators are normalized first. Then for each algorithm in each problem, the average of the indicators is calculated (with the same weight) and then considered as the performance of that algorithm in the given problem. These comparisons are shown in Figures 12–14 for the uncertainty levels of 0.2, 0.4 and 0.5. Regarding Figures 12–14, the NSGA-II and SFLA algorithms perform similarly to the optimal method to a large extent.

However, to solve the problem in a large size (4th instance), the proposed meta-heuristic algorithms were used because the ϵ -constraint method cannot provide accurate solutions for such problems. To determine which algorithm has a better performance in large-sized

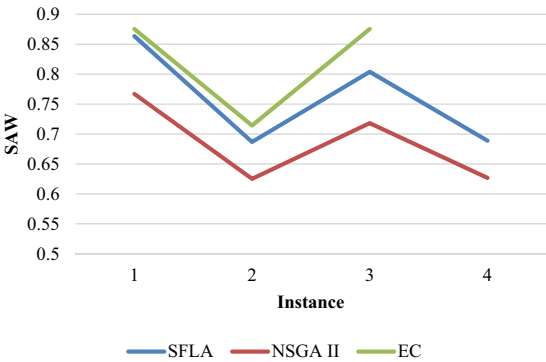


Figure 12.
Overall comparison of
the performance of EC,
NSGA-II and SFLA
methods at uncertainty
level of 0.2

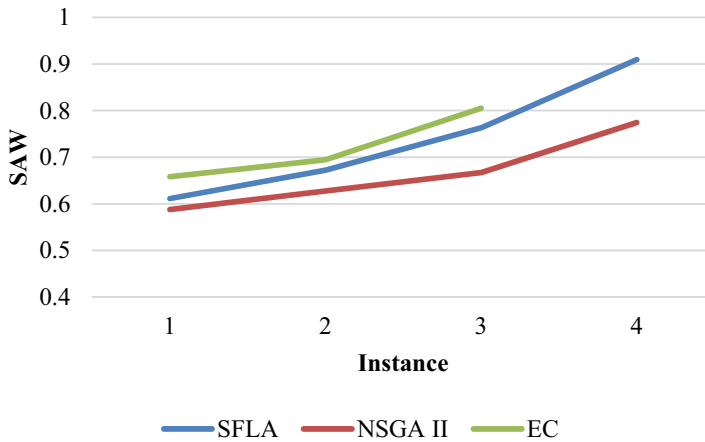


Figure 13.
Overall comparison of
the performance of EC,
NSGA-II and SFLA
methods at uncertainty
level of 0.4

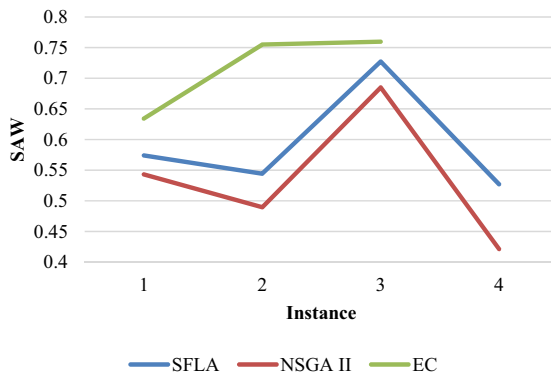


Figure 14.
Overall comparison of
the performance of EC,
NSGA-II and SFLA
methods at uncertainty
level of 0.5

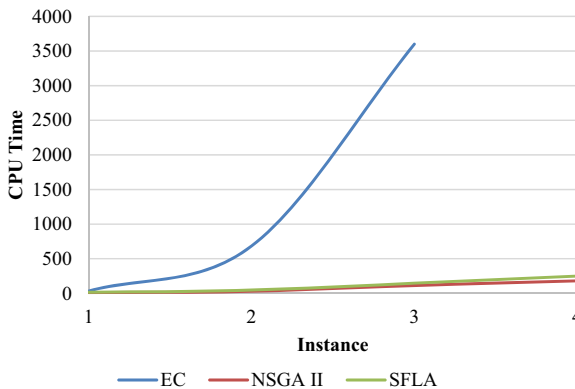


Figure 15.
Solution time of the
proposed solution
methods

instances, the values of the indicators are investigated in [Figures 12–14](#). These figures reveal that the SFLA algorithm has a much better performance in general.

[Figure 15](#) also shows the solution time for different instances. According to [Figure 14](#), increasing the dimensions of the problem leads to a significant increase in the solution time of the exact method. As a result, in the 4th instance, the ε -constraint method could not solve the problem in the given time limitation. On the other hand, the meta-heuristic algorithms were capable of solving the problem in a much shorter time. The NSGA-II algorithm requires less time to find the Pareto solutions compared to the SFLA algorithm.

5.3 Comparison of the deterministic and robust problems

This subsection provides an analytical comparison between the results obtained by the deterministic and robust problems. In fact, as a great managerial implication, it would be useful to study the effects of the uncertainty on the problem which may even make the problem infeasible due to resource shortages. To achieve a better perception of the effect of uncertain parameters and concerning the use of the RO approach to solve the problem, the results of the problems in deterministic conditions are compared the robust problem at an uncertainty level of 0.5 using the SFLA algorithm in [Table 11](#).

According to [Table 11](#), the first objective of the robust model is lower than the first objective in all of the deterministic instances, which is one of the benefits of RO. This indicates that in robust instances, the focus is on the reduction of departure maximum time from the given points in total, which can only be achieved by spending higher costs. In contrast to the first objective, the second objective in all of the robust instances is much more than the second objective of the deterministic instances, implying that having a robust problem and solution requires considering higher resources and subsequently, a higher total cost. Such an increase in cost is observable in the last four columns of [Table 11](#).

One of the major managerial achievements is the special attention paid to the uncertain conditions in the real world, which would lead to the changes in the solution space as well as the results. Robustness also has a direct impact on the service-providing facilities and equipment in the case of disasters and eventually the total costs. Most importantly, this impacts on the response time. This can be associated with increasing the costs in order to impose a subsequent reduction in response time.

5.4 Discussion on different statuses

This subsection provides a quick reference guide to the different statuses. To this end, [Table 12](#) is presented as a summary table showing the superior performance of each solution method.

According to [Table 12](#), it is concluded that the ε -constraint method is regarded as the best solver for small- and medium-sized instances and that SFLA is the best for large-sized instances.

5.5 Sensitivity analyses

In this subsection, the impacts of changing the transferring times; i.e. t_{ij} on the objective functions is studied using sensitivity analysis. In other words, t_{ij} can be regarded as a determinant parameter from the perspective of the operations managers. Hence, management wants to know how much this change may have a positive/negative effect on the objective functions. To this end, the fluctuation range of -20% , -10% , 0% , $+10\%$ and 20% are taken into account for t_{ij} . Then the obtained values of both objective functions were evaluated for the robust problem using the SFLA algorithm. The 4th instance was considered to conduct this sensitivity analysis. The results are represented in [Table 13](#) and [Figures 16–17](#).

Instance	First objective function (min)	Second objective function (\$)	Solving time (sec)	Number of established transfer stations	Number of established temporary hospitals	Number of ambulances	Number of helicopters
1 (Deterministic)	114.113	10081.49	12.26	1	1	1	1
1 (Robust)	53.54	34420.72	15.26	2	2	2	1
2 (Deterministic)	135.351	20202.70	34.1	2	3	4	1
2 (Robust)	29.95	57865.72	46.15	3	4	7	1
3 (Deterministic)	123.528	30478.42	141.54	3	3	10	2
3 (Robust)	41.66	65745.14	147.28	4	4	13	3
4 (Deterministic)	142.04	94205.46	208.65	5	8	16	4
4 (Robust)	106.05	133844.70	249.13	5	9	18	5

Table 11.
Comparison of
objective values
obtained by
deterministic and
robust problems

Evidently, the changes in transferring times have a significant impact on the 1st and 2nd objective functions. Creating a small increase in this parameter leads to a nonlinear increase in both objective functions. In other words, the objective functions have a direct relation with this parameter. The main reason for the increase in the objective functions against the increase of the parameter is because more resources are required to provide the relief services.

Table 12.
Performance
comparison of different
solution methods for
different instances and
indicators

Instance	NPS	SM	Indicators		CPU time	Superior solution method based on SAW
			MID	DM		
2	SFLA		EC	EC	NSGA-II	EC
	SFLA	EC	EC	EC	NSGA-II	EC
3	SFLA	EC	EC	EC	NSGA-II	EC
4	SFLA	SFLA	SFLA	SFLA	NSGA-II	SFLA

Table 13.
Sensitivity analysis
results of the 1st
objective function

Objective functions	Change percentage of t_{ij}				
	-20%	-10%	0%	+10%	+20%
Obj. 1	96.32	102.15	106.05	112.01	124.08
Obj. 2	125238.49	129702.94	133844.70	137112.48	147407.51

Figure 16.
The behavior of the 1st
objective function
against different
transferring times

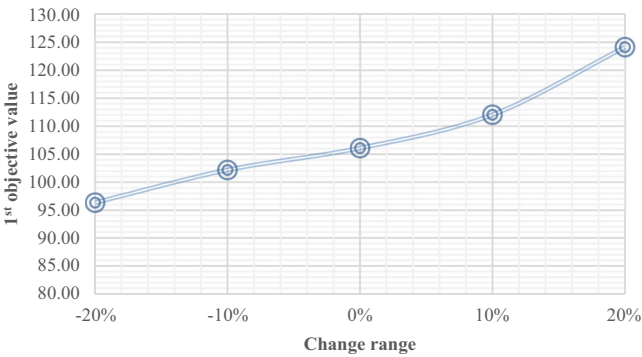
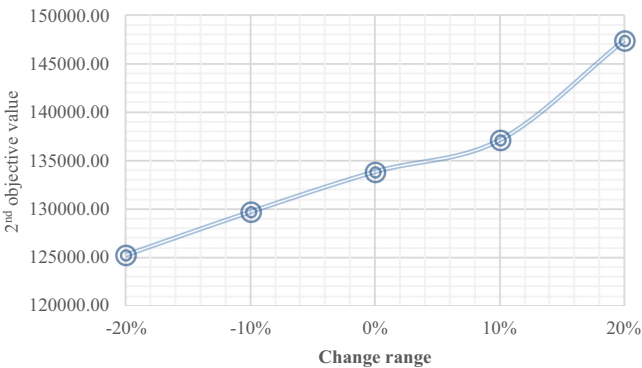


Figure 17.
The behavior of the
2nd objective function
against different
transferring times



Noticeably, by investigating these results, operations managers can implement sensitivity analysis as a useful tool in their decision-making process.

6. Conclusion

This paper focused on the investigation of relief planning in the case of urban disaster considering demand uncertainty. In this paper, there are injured patients with different levels of severity in different parts of the city and the relief system is equipped with ambulances and helicopters. Based on real conditions, the helicopters are responsible for transferring the heavily injured patients and the ambulances are responsible for providing relief services at the site and for transferring the patients to the medical centers. The proposed work aimed to plan the location and routing decisions as a robust LRP. In the first phase of decision making, the optimal location of the emergency facilities, as well as those of temporary hospitals, were determined. In the second phase, the optimal routes for transferring the patients along different points of the network were determined in order to accomplish the process of providing services in the shortest possible time. To overcome the uncertainty, the RO approach was used. Then, solving the problem and the verification and validation of the proposed model were both performed using the GAMS software. Subsequently, the SFLA algorithm was developed to solve the problem approximately. The performance of the proposed algorithm was compared to the ϵ -constraint methods in small-sized instances and the NSGA-II algorithm in large-sized instances. After solving the problem, it was revealed that the proposed algorithms had high efficiency in terms of providing a reasonable solution time. In general, the SFLA algorithm had a better performance than the NSGA-II algorithm in 4 indicators; i.e. NPS, SM, MID and DM. Accordingly, SFLA covered a wider range of solutions with better quality and it also generated a more uniform front. However, the solution time of the NSGA-II algorithm was better. Furthermore, one of the considerable outputs was to obtain a better first objective and a worse second objective in the robust problem compared to the deterministic problem. This indicated that in the robust problem, the focus was totally put on reducing the maximum time of departure from the given points which could be achieved but with higher costs. A suggestion for further works is to consider the issue of facility reliability and the route traffic as well as the use of other meta-heuristic algorithms to be compared with the SFLA. Moreover, different rescue teams with different levels of service-providing may be defined in the problem to augment the efficiency, effectiveness and equity measures.

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Tables A1–A3 represent the calculated values for the fronts obtained by the two algorithms for all these four problems at various uncertainty levels.

Indicators instance	NPS		SM		MID		DM			
	SFLA	NSGA II	EC	SFLA	EC	SFLA	EC	SFLA	NSGA II	EC
1	9	7	–	0.98	0.92	0.78	0.91	1.21	1.13	1.25
2	22	15	–	1.14	1.07	1.24	1.29	1.12	1.04	1.16
3	31	28	–	0.67	0.69	0.94	1.11	0.84	0.71	0.95
4	49	41	–	2.45	2.61	1.47	1.61	1.22	1.19	–

Table A1.
Mean values of the
indicators obtained for
three algorithms at
uncertainty level of 0.2

Indicators Instance	NPS			SM			MID			DM		
	SFLA	NSGA II	EC	SFLA	NSGA II	EC	SFLA	NSGA II	EC	SFLA	NSGA II	EC
1	8	7	–	2.21	2.33	2.13	0.73	0.74	0.68	0.89	0.83	0.98
2	19	16	–	2.08	2.1	2.02	0.87	0.92	0.83	1.33	1.2	1.36
3	44	26	–	0.74	0.92	0.72	0.79	0.84	0.75	0.71	0.64	0.79
4	63	58	–	1.13	1.24	–	0.65	0.91	–	1.44	1.27	–

Table A3.
Mean values of the
indicators obtained for
three algorithms at
uncertainty level of 0.5